

12.5

DATA SET

DS1219

12. What to Buy? A marketing research experiment was conducted to study the relationship between the length of time necessary for a buyer to reach a decision and the number of alternative package designs of a product presented. The products were identical except for the package design. The length of time necessary to reach a decision was recorded for 15 participants in the marketing research study.

Length of Decision Time, y (sec)	5, 8, 8, 7, 9	7, 9, 8, 9, 10	10, 11, 10, 12, 9
Number of Alternatives, x	2	3	4

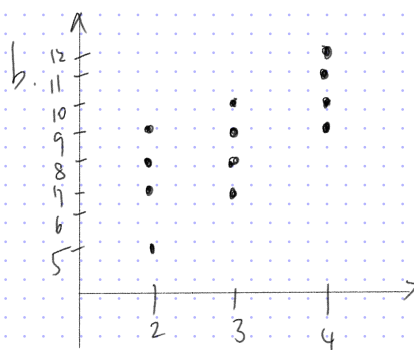
- Find the least-squares line appropriate for these data.
- Plot the points and graph the line as a check on your calculations.
- Calculate s^2 .
- Do the data present sufficient evidence to indicate that the length of decision time is linearly related to the number of alternative package designs? (Test at the $\alpha = .05$ level of significance.)
- Find the approximate p -value for the test and interpret its value.
- If they are available, examine the diagnostic plots to check the validity of the regression assumptions.
- Estimate the average length of time necessary to reach a decision when three alternatives are presented, using a 95% confidence interval.

$$A. \hat{y} = a + bx \quad \bar{y} = 8.8 \quad \bar{x} = 3$$

$$b = \frac{S_{xy}}{S_{xx}} = \frac{\sum xy - n\bar{x}\bar{y}}{\sum x^2 - n\bar{x}^2} = 1.5$$

$$a = \bar{y} - b\bar{x} = 8.8 - 1.5 \times 3 = 4.3$$

$$\hat{y} = 4.3 + 1.5x$$



$$d. H_0: \beta = 0, H_a: \beta \neq 0$$

$$\alpha = 0.05$$

$$t_{STAT} = \frac{b}{\sqrt{\frac{MSE}{S_{xx}}}} \sim t_{n-2}$$

$$t^* = \frac{1.5}{\sqrt{\frac{1.531}{10}}} = 3.834$$

$$RR = \{t \mid |t| > t_{0.025, 13}\} = \{t \mid |t| > 2.16\}$$

$$3.834 > 2.16 \Rightarrow \text{Reject } H_0$$

$\Rightarrow$ there are significant linear relationship between x, y

$$e. 2p(x > 3.834) = 0.021 < 0.05 \Rightarrow \text{Reject } H_0$$

c.

ANOVA	SS	df	MS	F
SSR	22.5	1	22.5	14.696
SSE	19.9	13	1.531	
SS	42.4	14		

$$SSR = \frac{S_{xy}^2}{S_{xx}} = 22.5$$

$$SS = S_{yy} = 42.4$$

$$SSE = 42.4 - 22.5 = 19.9$$

$$s^2 \approx MSE = 1.531$$

$$g. \hat{y} = 4.3 + 1.5x$$

$$95\% \text{ C.I. } \hat{y} \pm t \sqrt{MSE \left(1 + \frac{1}{n} + \frac{(x_p - \bar{x})^2}{S_{xx}} \right)}$$

$$x=2: \hat{y} = 4.3 + 3 = 7.3$$

$$t_{0.025, 13} = 2.16$$

C.I.

$$x=3: \hat{y} = 8.8$$

$$x=2: 2.16 \cdot \sqrt{1.531 \left(1 + \frac{1}{15} + \frac{(2-3)^2}{10} \right)} = 2.8868 \quad (4.4132, 10.1868)$$

$$x=4: \hat{y} = 10.3$$

$$x=3: 2.16 \cdot \sqrt{1.531 \left(1 + \frac{1}{15} + \frac{(3-3)^2}{10} \right)} = 2.16029 \quad (6.0397, 11.5603)$$

$$x=4: 2.16 \cdot \sqrt{1.531 \left(1 + \frac{1}{15} + \frac{(4-3)^2}{10} \right)} = 2.8868 \quad (7.4132, 13.1868)$$